



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Case Study : 5

1. Title: Study of Arrays and Loops in JavaScript with Maximum and Minimum Value Calculation.

2. Software/Tools Required: HTML, VS Code.

3. Theory: Arrays in JavaScript are used to store multiple values in a single variable. Array methods such as `push()`, `pop()`, `shift()`, `unshift()`, `splice()`, and `slice()` are used to add, remove, and extract elements from an array. Methods like `map()`, `filter()`, `reduce()`, and `forEach()` help in processing and manipulating array elements. In this case study, an array of numbers is created, and JavaScript is used to find the maximum and minimum values using `Math.max()` and `Math.min()`.

4. Output:

```
Original Array: 10,20,30,40,50

After push(60): 10,20,30,40,50,60
After pop(): 10,20,30,40,50
After unshift(5): 5,10,20,30,40,50
After shift(): 10,20,30,40,50
After splice(): 10,20,35,40,50
After slice(1,4): 20,35,40

forEach(): 10 20 35 40 50
map() - Doubled Array: 20,40,70,80,100
filter() - Numbers > 20: 35,40,50
reduce() - Sum: 155

Case Study
Array: 10,20,35,40,50
Maximum Value: 50
Minimum Value: 10
```



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Conclusion : In this experiment, we successfully studied arrays and loops in JavaScript. We implemented different array methods to add, remove, and manipulate elements. We also used `map()`, `filter()`, `reduce()`, and `forEach()` for array processing. The case study helped us find the maximum and minimum values from an array of numbers and understand the practical use of JavaScript array methods.